

Chapter 87

Particle Systems

This chapter is designed to give you a brief introduction to the creation of fully 3D particle systems, one of Fusion’s most powerful features.

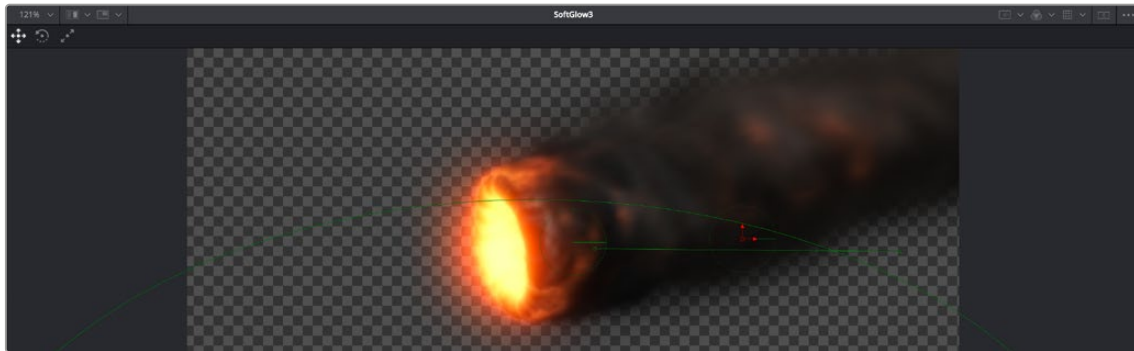
Once you understand these basics, for more information on each particle system node that’s available, see *Chapter 114, “Particle Nodes,”* in the DaVinci Resolve Reference Manual or Chapter 54 in the Fusion Reference Manual.

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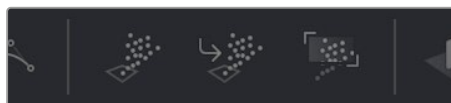
Introduction to Particle Systems

Particle systems are computer simulations that use customizable rules to automatically generate and animate large numbers of elements to simulate smoke, dust, fire, leaves, sparks, or any other animated system of shapes. As Fusion is a full-featured 3D compositing environment, particle systems can be created in 2D or 3D, which makes them incredibly flexible and capable of producing all kinds of visual effects or abstract animated content for use in motion graphics.



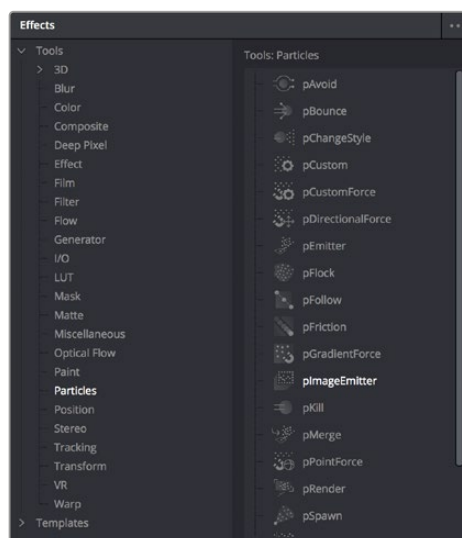
A 3D particle system, also created entirely within Fusion

The three most fundamental nodes required for creating particle systems are found on the toolbar. As with the 3D nodes to the right, these are arranged, from left to right, in the order in which they must be connected to work, so even if you can't remember how to hook up a simple particle system, all you need to do is click the three particle system nodes from left to right to create a functional particle system.



The pEmitter, pMerge, and pRender particle system nodes available from the toolbar

However, these three nodes are only the tip of the iceberg. Opening the Particle category in the Effects Library reveals many, many particle nodes designed to work together to create increasingly complex particle interactions.

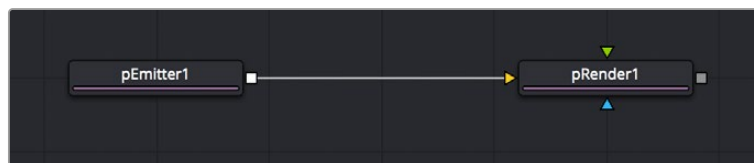


A sample of the nodes available in the Particles bin of the Effects Library

All particle nodes begin with the letter “p,” and they’re designed to work together to produce sophisticated effects from relatively simple operations and settings. The next section shows different ways particle nodes can be connected to produce different effects.

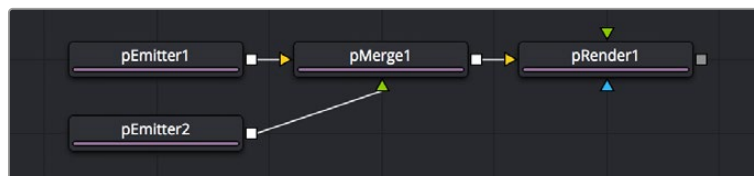
Anatomy of a Simple Particle System

The simplest particle system you can create is a pEmitter node connected to a pRender node. The pEmitter node includes the core controls for creating various kinds of particles in different ways, while the pRender node is required to render a 2D or 3D result that can be composited with other scenes within your composition.



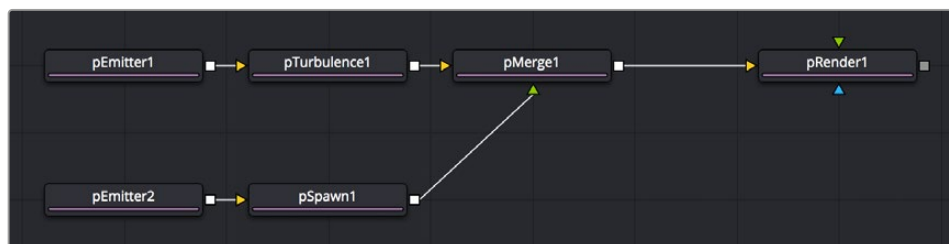
The minimum node tree required to create a simple particle system

If your needs are more complicated, you can combine two or more pEmitter nodes using a pMerge node (the particle system version of a Merge node), to create compound particle systems where multiple types of particles combine with one another to create a result.



Compositing two pEmitter nodes to create a compound particle system, combining two kinds of particles together

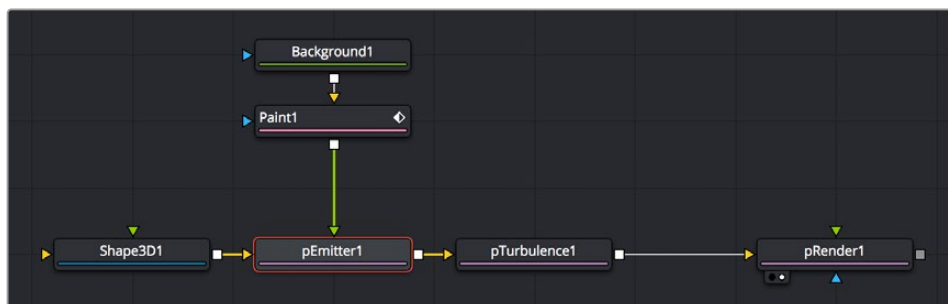
If you’re trying to create particle systems with more natural effects, you can add “forces” to each emitter. These forces are essentially physics or behavioral simulations that automatically cause the particles affected by them to be animated with different kinds of motion, or to be otherwise affected by different objects within scenes.



Customizing the effect of pEmitter nodes using different forces to add complexity to the particle animation

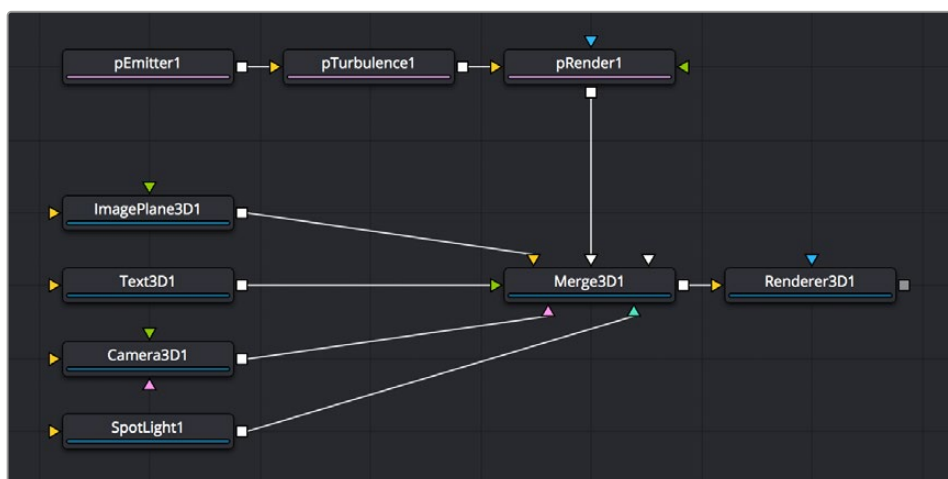
You can also attach the following types of nodes to a pEmitter node to deeply customize a particle system:

- Attach a 2D image to a pEmitter node to create highly customized particle shapes. Make sure your image has an appropriate alpha channel.
- Attach a Shape3D or other 3D geometry node to a pEmitter node to create a more specific region of emission (by setting Region to Mesh in the Region tab).



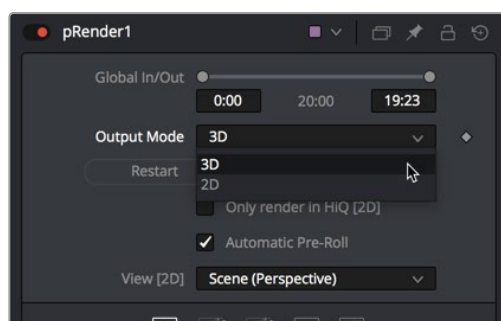
Customizing pEmitter nodes using mesh geometry to define regions and 2D images to define particle shape

The above examples assume that you'll output 2D renders to combine into the rest of a 2D composition. However, because particle systems are fully 3D, you also have the option of outputting your particle system in such a way as to be used from within other 3D scenes in your composition.



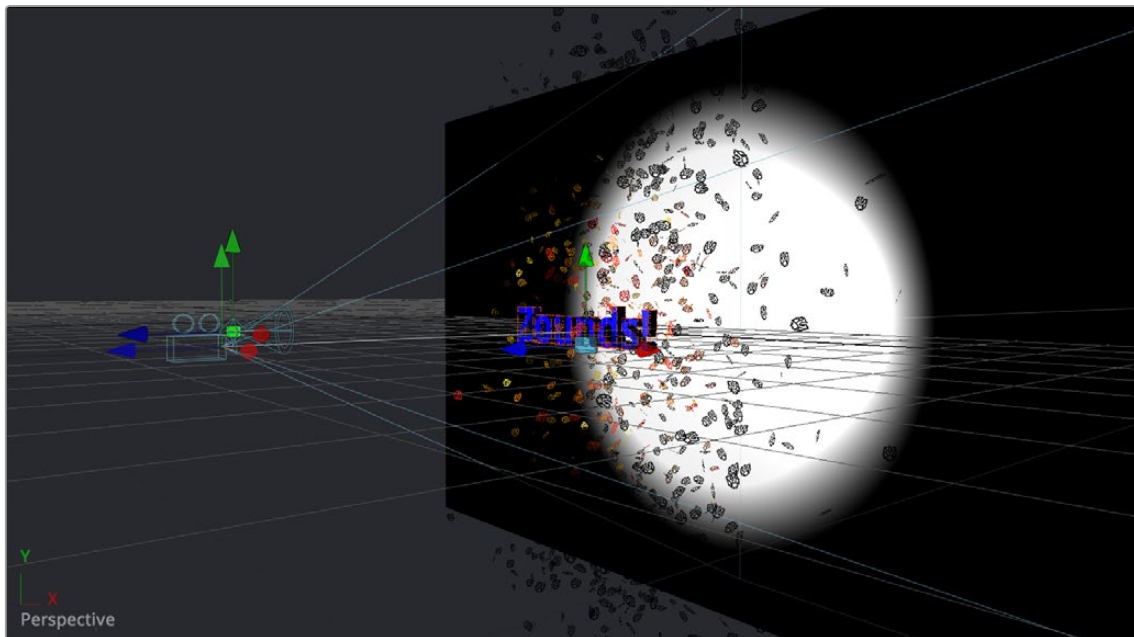
Connecting a particle system to a Merge3D node so the particles are subject to lighting and shadows within a 3D scene

The Output Mode of the pRender node, at the very top of the controls exposed in the Inspector, can be set to either 2D or 3D, depending on whether you want to combine the result of the particle system with 2D layers or with objects in a 3D scene.



Choosing whether a particle system's output is 2D or 3D in the pRender node's Inspector controls

If you connect a pRender node to a Merge3D node, the Output Mode is locked to 3D, meaning that 3D geometry is output by the pRender node for use within the Merge3D node's scene. This means that the particles can be lit, they can cast shadows, and they can interact with 3D objects within that scene.

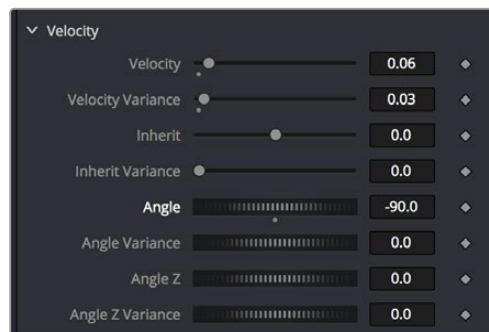


The result of using a particle system within a 3D scene

NOTE: Once you set the pRender node to either 2D or 3D and make any change to the nodes in the Inspector, you cannot change the output mode.

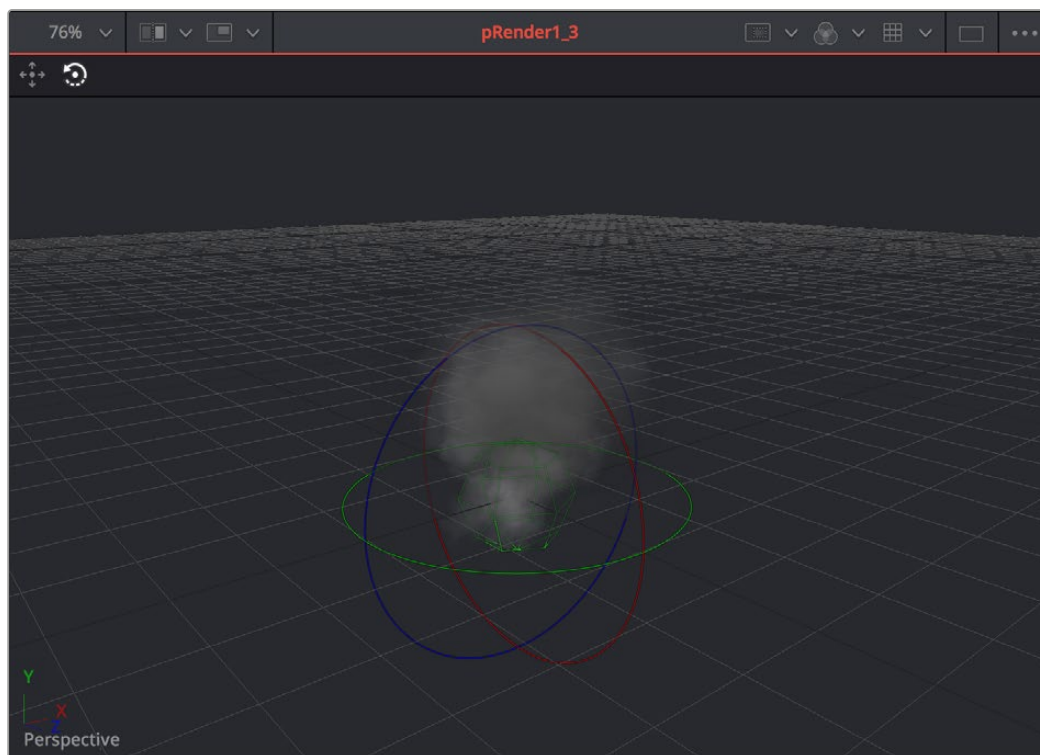
Particle System Distribution

To adjust the distribution of particles being emitted, select the pEmitter node to expose its controls in the Inspector, then open the Velocity controls in the Controls tab, and use the Angle, Angle Variance, Angle Z, and Angle Z Variance controls to adjust the direction and width over which particles are emitted. All these controls can be animated.



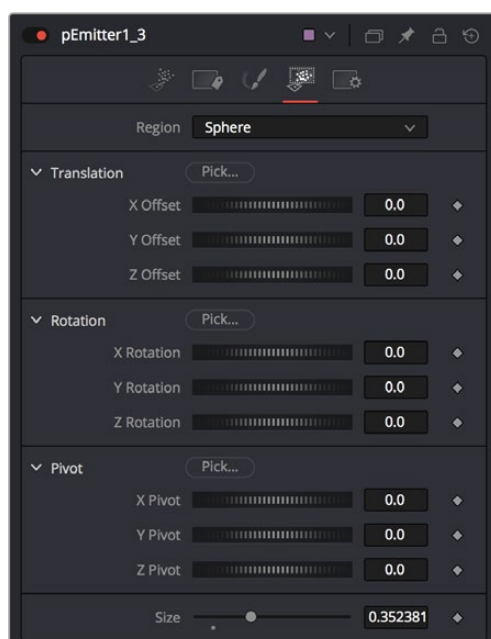
A pEmitter node's Velocity Angle and Angle Variance controls let you adjust the direction and width of particle distribution

Particle systems can be positioned and rotated by loading the pEmitter nodes that generate particles into a viewer and using the onscreen 3D position and Rotation controls provided to move the particle system around.



A pEmitter node loaded into the viewer with the rotation onscreen controls enabled

Alternatively, you can use the controls of the pEmitter's Region tab in the Inspector to adjust Translation, Rotation, and Pivot. All these controls can be animated.



A pEmitter node's Region controls open in the Inspector

Particle Nodes Explained by Type

This section introduces the four types of particle system nodes available in the Effects Library.

Emitters

pEmitter nodes are the source of all particles. Each pEmitter node can be set up to generate a single type of particle with enough customization so that you'll never create the same type of particle twice. Along with the pRender node, this is the only other node that's absolutely required to create a particle system.

pEmitter nodes have four parameters tabs:

Controls: The primary controls governing how many particles are generated (Number), how long they live (Lifespan), how fast they move (Velocity) and how widely distributed they are (Angle and Angle Variance), their rotation (Rotation Mode with X, Y, and Z controls), and whether there's spin (Spin X, Y, and Z controls). For each parameter of particle generation, there's an accompanying Variance control that lets you make that parameter less uniform and more natural by introducing random variation.

Sets: This tab contains settings that affect the physics of the particles emitted by the node. These settings do not directly affect the appearance of the particles. Instead, they modify behaviors such as velocity, spin, quantity, and lifespan.

Style: While the Controls tab has a simple control for choosing a color for particles, the Style tab has more comprehensive controls including color variance and Color Over Life controls. Additionally, size controls including Size Over Life, fade controls, and blur controls let you create sophisticated particle animations with a minimum of adjustments, while Merge controls give you an additional level of control over how overlapping particles combine visually. A set of controls at the bottom lets you choose how animated effects are timed.

Region: The Region tab lets you choose what kind of geometric region is used to disperse particles into space and whether you're emitting particles from the region's volume or surface. The Winding Rule and Winding Ray Direction controls determine how the mesh region will handle particle creation with geometric meshes that are not completely closed, as is common in many meshes imported from external applications. Tweaking these last parameters is common when using imported mesh geometry as a region for emitting particles, since even geometry that appears closed will frequently appear to "leak" particles thanks to improperly welded vertices.

Forces

Many of the particle nodes found in the Particles bin of the Effects Library are "forces" that enhance a particle simulation by simulating the effect of various forces acting upon the particles generated by an emitter.

Some forces, including pDirectionalForce, pFlock, pFriction, pTurbulence, and pVortex, are rules that act upon particles without the need for any other input. These are simply "acts of nature" that cause particles to behave in different ways.

Other forces, such as pAvoid, pBounce, pFollow, and pKill, work in conjunction with 3D geometry in a scene such as shapes or planes to cause things to happen when a particle interacts or comes near that geometry. Note that some of the particles described previously can also use geometry to direct their actions, so these two categories of forces are not always that clear-cut.

Compositing

The pMerge node is a simple way to combine multiple emitters so that different types of particles work together to create a sophisticated result. The pMerge node has no parameters; you simply connect emitters to it, and they're automatically combined.

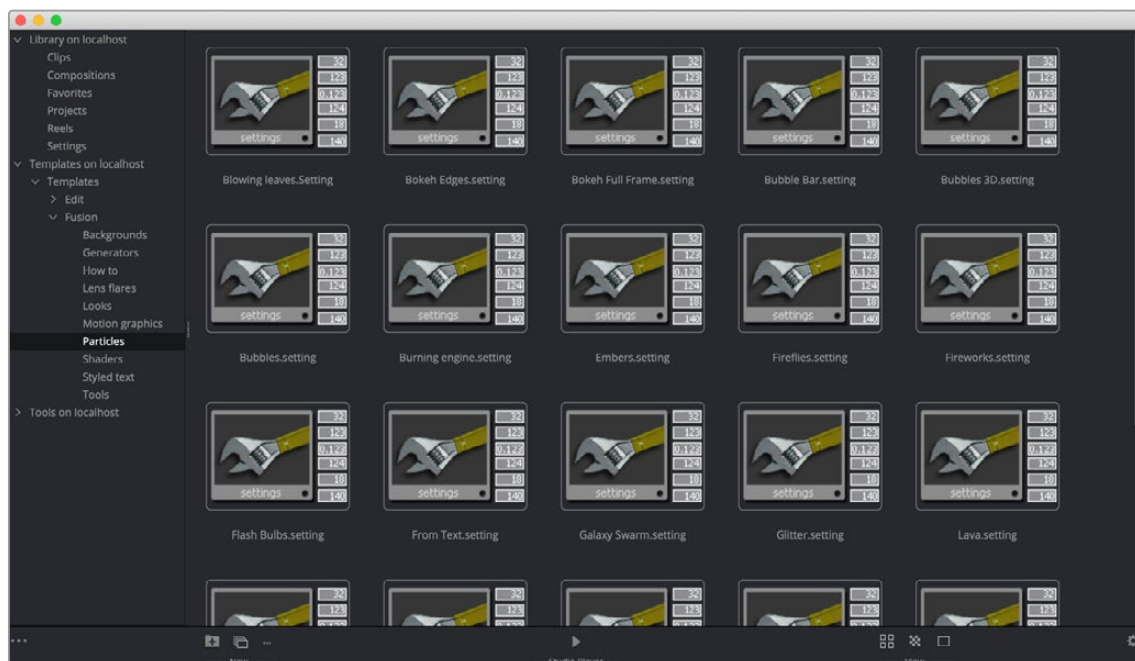
Rendering

The pRender node is required whether you're connecting a particle system's output to a 2D Merge node or to a Merge3D node for integration into a 3D scene. Along with the pEmitter node, this is the only other node that's absolutely required to create a particle system.

- **Controls:** The main controls that let you choose whether to output 2D or 3D image data, and whether to add blur or glow effects to the particle systems, along with a host of other details controlling how particles will be rendered.
- **Scene:** These controls let you transform the overall particle scene all at once.
- **Grid:** The grid is a helpful, non-rendering guide used to orient 2D particles in 3D space. The grid is never output in renders. The width, depth, number of lines, and grid color can be set using the controls found in this tab.
- **Image:** Controls the output of the pRender node, with controls over the process mode, resolution, and color space settings of the output.

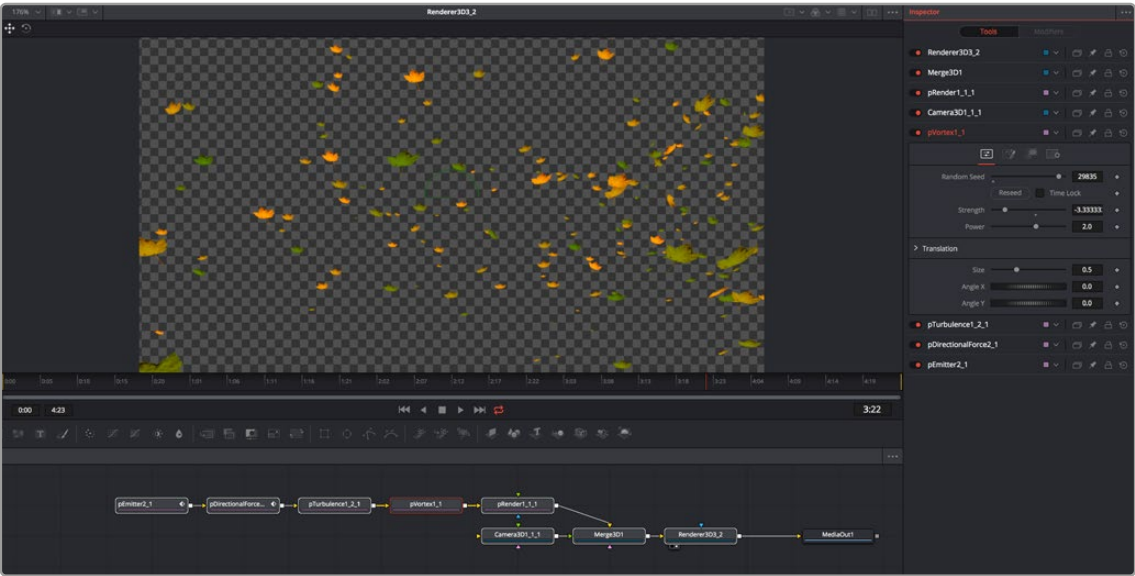
Example Particle Systems

The Templates category in the Inspector in the Fusion page of DaVinci Resolve or in the Bins window in Fusion Studio includes over 20 different examples of particle systems creating a variety of effects. One of the best ways of learning how to create and customize particle systems is to open these and investigate how they're made.



Different particle system presets in the Templates category of the Bins window in Fusion Studio

Simply drag and drop any of the particle presets into the Node Editor, load the last node into the viewer, and you'll see how things are put together.



The Blowing Leaves preset from the Templates category